

STAT6001

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Assumptions: Linearity
Normality (of e_i)
Homogeneity ($e_i \perp x_i$)
Independence (of e_i)

Linear Models

$$y = X\beta + e \quad \left(\begin{array}{l} e_i \stackrel{iid}{\sim} N(0, \sigma^2) \\ y_i \stackrel{iid}{\sim} N(\mu_i, \sigma^2), \mu_i = x_i^T \beta \end{array} \right)$$

$$S(\beta) = \sum e_i^2 = \sum (y_i - x_i^T \beta)^2 = (y - X\beta)^T (y - X\beta)$$

unbiased if y_i normal $\rightarrow \frac{\partial S}{\partial \beta} = 2X^T X \beta - 2X^T y = 0 \Rightarrow \hat{\beta} = (X^T X)^{-1} X^T y$

$$\hat{\beta} \sim N(\beta, \sigma^2 (X^T X)^{-1})$$

$$\hat{y} = X\hat{\beta} = Hy \Rightarrow H = X(X^T X)^{-1} X^T \quad \left(\begin{array}{l} \text{tr}(H) = p \\ H^T H = H \\ H^T = H \end{array} \right)$$

$$\hat{e} = y - \hat{y} = (I - H)y, \quad \left(\begin{array}{l} X^T \hat{e} = 0 \quad (X^T X \beta - X^T y = 0) \\ \sum e_i = 0 \end{array} \right)$$

% Variance explained,
low = don't trust
model inferences

$$RSS = \hat{e}^T \hat{e}, \quad E(\hat{e}^T \hat{e}) = E(y^T (I - H)y) = (N - p) \sigma^2$$

$$\Rightarrow \hat{\sigma}^2 = \frac{RSS}{N - p}, \quad \left(\frac{RSS}{\sigma^2} \sim \chi^2_{N-p} \right)$$

WLS: $e_i \sim N(0, V_{ii})$, V diagonal $W_i = \frac{1}{\text{Var}(y_i)}$

$$S(\beta) = \sum w_i (y_i - \hat{y}_i)^2 = (y - X\beta)^T V^{-1} (y - X\beta)$$

$$\Rightarrow \hat{\beta} = (X^T V^{-1} X)^{-1} X^T V^{-1} y$$

$$\left(\begin{array}{l} E(\hat{\beta}x) = \hat{\beta} E(x) \\ V(\hat{\beta}x) = \hat{\beta} V(x) \hat{\beta}^T \end{array} \right)$$

$$R^2 = \frac{\sum (\hat{y}_i - \bar{y})^2}{\sum (y_i - \bar{y})^2} = 1 - \frac{RSS}{CTSS}$$

$\frac{N-1}{2} > 2$

$$r_i = \frac{\hat{e}_i}{\hat{\sigma} \sqrt{1 - h_{ii}}}$$

$$(\hat{e}_i \sim N(0, \sigma^2 (I - H))) \leftarrow (\Rightarrow \hat{e}_i \perp \hat{e}_j, \text{ as } \sum e_i = 0)$$

$$\begin{aligned} V(\hat{e}) &= V(y - \hat{y}) = V((I - H)y) \\ &= (I - H)V(y)(I - H)^T \\ &= \sigma^2 (I - H) \end{aligned}$$

Regularisation

$$\hat{\beta}_{RR} = (X^T X + \lambda I)^{-1} X^T y, \quad E(\hat{\beta}_{RR}) = X^T X (\lambda I + X^T X)^{-1} \beta$$

• Partial vs total regression coeff

• t-test: $\hat{\beta}_j \sim N(\beta_j, \sigma^2 V_j)$, $V_j = [(X^T X)^{-1}]_{jj}$

$$\Rightarrow \frac{\hat{\beta}_j - \beta_j}{se(\hat{\beta}_j)} \sim t_{N-p} \quad (se(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 V_j})$$

$\hat{\beta}_j \pm t_{N-p, \frac{1}{2}\alpha}$ confidence interval $(1-\alpha)\%$

• Robust Regression:

(+ve, symmetric, $\rho(0)=0$)

$$S(\beta) = \sum \rho\left(\frac{e_i(\beta)}{\sigma}\right)$$

($\psi = \rho'$)
(bounded ψ = bounded effect of large e_i)

(Also L₁ & Least Median)

analogous to WLS, except $\underline{W}$ not constant

$$\hat{\beta}_{(k+1)} = (\underline{X}^T \underline{W}^{(k)} \underline{X})^{-1} \underline{X}^T \underline{W}^{(k)} \underline{y}$$

↑
iterate to convergence

$$\begin{aligned} W_i^{(k)} &= \frac{\psi(u_i^{(k)})}{u_i^{(k)}} \\ u_i^{(k)} &= (y_i - \hat{y}_i^{(k)}) / \hat{\sigma}^{(k)} \\ \hat{y}_i^{(k)} &= \underline{x}_i^T \hat{\beta}^{(k)} \\ \hat{\sigma}_{MAD}^{(k)} &\propto \text{MED}\{|\hat{e}_i| - \text{MED}\{\hat{e}_j\}\} \end{aligned}$$

• Variable Selection

• Backwards & Forwards elimination ($p=d+1$)

$$AIC = -2\hat{\lambda}(\text{model}) + 2p = N \log \frac{RSS}{N} + 2p \quad \left(\lambda(\text{model}) = -\frac{N}{2} \log \hat{\sigma}^2 - \frac{RSS}{2\hat{\sigma}^2} \right)$$

$$\hat{\sigma}^2 = \frac{RSS}{N}$$

• LOO-CV

• Lasso (standardize $x_{ij} = \frac{z_{ij} - \bar{z}_j}{s_j}$)

Generalized Linear Models

$$\eta_i = \underline{x}_i^T \underline{\beta}$$

link function $\rightarrow g(\mu_i) = \eta_i, \mu_i = E(y_i), y_i \sim \text{Exp fam}(\theta, \phi)$

default for R $\rightarrow$

$y_i \sim N(\mu_i, \sigma^2) \Rightarrow g(\mu) = \mu$	] Canonical links, $g(\mu) = \theta$
$y_i \sim \text{Poisson}(\mu_i) \Rightarrow g(\mu) = \log(\mu)$	
$y_i \sim \text{Binomial}(n, p) \Rightarrow g(\mu) = \logit(\mu) = \log \frac{\mu}{1-\mu}$	

$$\ell(\underline{\beta}) = \sum \log P(y_i)$$

$$\frac{\partial \ell}{\partial \beta_j} = \sum \left(\frac{y_i - \mu_i}{v(\mu_i)} \right) \frac{d\mu_i}{d\eta_i} x_{ij} = 0$$

Estimate $\underline{\beta}$ $\rightarrow$ Fisher iterations

$$\Rightarrow (\underline{X}^T \underline{W} \underline{X})^{(s-1)} \hat{\underline{\beta}}^{(s)} = (\underline{X}^T \underline{W} \underline{z})^{(s-1)}$$

$$W_{ii} = \frac{1}{v(\mu_i)} \left(\frac{d\mu_i}{d\eta_i} \right)^2$$

$$\underline{z}^{(s)} = \underline{\eta}_i + (y_i - \mu_i^{(s)}) \left(\frac{d\mu_i}{d\eta_i} \right)^{(s)}$$

Fisher Info Matrix

$$\rightarrow \underline{I}_{\beta_j \beta_j} = E \left(\frac{\partial \ell}{\partial \beta_j} \cdot \frac{\partial \ell}{\partial \beta_j} \right), \underline{I} = \underline{X}^T \underline{W} \underline{X} / \phi (= \text{negative inverse of Hessian})$$

$$\hat{\underline{\beta}} \sim N(\underline{\beta}, \underline{I}^{-1}) \Rightarrow \frac{\hat{\beta}_j - \beta_j}{\text{se}(\hat{\beta}_j)} \sim N(0, 1)$$

Deviance:

$$D^* = 2 [\ell(\hat{\underline{\beta}}_{\text{sat}}) - \ell(\hat{\underline{\beta}})] \phi \quad \left(\ell(\hat{\underline{\beta}}_{\text{sat}}) \text{ is likelihood when } \hat{\mu}_i = y_i \right)$$

$$D^* = D / \phi \sim \chi^2_{N-p}$$

likelihood test of H_0

$$\rightarrow D_0^* - D^* \sim \chi^2_{p-q}, \quad (D_0^* = \text{standardized deviance under } H_0: p-q \text{ params all 0})$$

$$D_{M_1}^* - D_{M_2}^* \sim \chi^2_{p_1 - p_2}$$

Estimate ϕ

$$\rightarrow \chi^2 = \sum \frac{(y_i - \hat{\mu}_i)^2}{v(\hat{\mu}_i)}, \quad \chi^2 / \phi \sim \chi^2_{N-p} \Rightarrow \hat{\phi} = \frac{\hat{\chi}^2}{N-p}$$

D

Exponential Family

$$f(y; \theta, \phi) = \exp \left[\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi) \right]$$

	θ	ϕ	$b(\theta)$	$c(y, \phi)$
Poisson (μ)	$\log \mu$	1	e^θ	$-\log y!$
Bin (n, π)	$\log \pi$	1	$n \log(1 + e^\theta)$	$\log \binom{n}{y}$
$N(\mu, \sigma^2)$	μ	σ^2	$\frac{1}{2} \theta^2$	$-\frac{1}{2} (y/\phi + \log(2\pi\phi))$

For GLM, Canonical link is $g(\mu) = \theta$

$$E(Y) = b'(\theta), \quad \text{Var}(Y) = b''(\theta) a(\phi) = V(\mu) a(\phi)$$

$\forall$

Generalized Additive Models

$$\eta = \underline{x}^T \underline{\theta} + \sum_j f_j(x_j) \quad (\text{instead of } \eta = \underline{x}^T \underline{\beta} \text{ for GLM})$$

f_j are smooth functions, eg splines

Spline: weighted sum of basis functions, $f_j(x_j) = \sum \beta_{jk} b_{jk}(x_j)$

• λ = smoothing parameters; punish excessively

wiggly fits
• estimate parameters w/ fixed λ by maximum likelihood

Linear Models

$$Y_i = \vec{x}_i^T \vec{\beta} + e_i, \quad i=1 \dots N, \quad \vec{x}_i \in D^m$$

- often assume $e_i \stackrel{iid}{\sim} N(0, \sigma^2)$ ($y_i \stackrel{iid}{\sim} N(x_i^T \vec{\beta}, \sigma^2)$) $D=R \Rightarrow$ multiple linear regression
- components of $\vec{x}$ can be transforms of the explanatory variable, eg $\vec{x}_i = \begin{pmatrix} 1 \\ u_i^2 \\ \log(u_i) \end{pmatrix}$ $D=\text{discrete} \Rightarrow$ analysis of variance model
 $D=\text{mix} \Rightarrow$ general linear regression

• useful if model assumptions (linearity, normality) violated for original variables

• or multiplicative model $Z_i = u_{i1} u_{i2} \dots u_{im} v_i$, take logs $\Rightarrow$
 $\log Z_i = \log u_{i1} + \log u_{i2} + \dots + \log v_i + e_i$

• or variable interactions, $\vec{x}_i = (1, u_i, v_i, u_i v_i)^T$

• or comparison of group means

$$Y_{ij} = \mu_i + e_{ij} \quad i=1 \dots I \quad j=1 \dots N_i$$

$$= x_{ij1} \mu_1 + x_{ij2} \mu_2 + \dots + e_{ij}$$

$$Y_{ij} = \vec{x}_{ij}^T \vec{\mu} + e_{ij}$$

where $\vec{x}_{ijk} = \begin{cases} 1 & \text{if } i=k \\ 0 & \text{otherwise} \end{cases}, \vec{x}_{ij} \in \{0, 1\}^I$

• Inference:

• Parameter estimation:

• Least Squares:

$$\hat{\beta} = \arg \min_{\beta} S(\beta), \quad S(\beta) = \sum_i e_i^2 = \sum_i (Y_i - \vec{x}_i^T \beta)^2$$

$$= (\vec{Y} - X\beta)^T (\vec{Y} - X\beta)$$

$$\Rightarrow X^T X \hat{\beta} = X^T \vec{Y}$$

if $p < N$, $X^T X$ invertible $\Rightarrow \hat{\beta} = (X^T X)^{-1} X^T \vec{Y}$

• $E(\hat{\beta}) = \vec{\beta}$, $V(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$ (if e_i independent, w/ var σ^2)

$\hat{\beta} \sim N(\vec{\beta}, \sigma^2 (X^T X)^{-1})$ if $e_i \stackrel{iid}{\sim} N(0, \sigma^2)$

• Maximum likelihood estimation

$$L(\vec{\beta}, \sigma^2) = \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^N \exp \left[-\frac{S(\vec{\beta})}{2\sigma^2} \right]$$

max of $L(\vec{\beta}, \sigma^2)$ for given σ equivalent to min $S(\vec{\beta})$

• Estimation of σ^2

• Residuals $\hat{e} = \vec{y} - X\hat{\beta}$, $\hat{\beta} = (X^T X)^{-1} X^T \vec{y}$
 $= (I_N - H) \vec{y}$, $H = X(X^T X)^{-1} X^T$

Residual Sum of Squares $\rightarrow RSS = \hat{e}^T \hat{e}$

$$E(RSS) = E(\hat{e}^T \hat{e}) = E(\vec{y}^T (I_N - H) \vec{y}) = (N-p)\sigma^2$$

$$\Rightarrow \hat{\sigma}^2 = \frac{RSS}{N-p} \quad (\text{if } e_i \sim N(0, \sigma^2), \text{ IID} \Rightarrow \frac{RSS}{\sigma^2} \sim \chi^2_{N-p})$$

diagonal, i.e. independent errors

• Weighted Least Squares Estimation

• use when error variance not constant $\leftarrow e_i \sim N(0, V)$
 • minimise $S(\vec{\beta}) = \sum w_i (y_i - \vec{x}_i^T \vec{\beta})^2$ ($w_i \uparrow = \text{more reliable measurements}$)
 ($w_i \uparrow = e_i \text{ lower variance}$)

if $w_i = \frac{1}{\text{Var}(y_i)}$, $S(\vec{\beta}) = (\vec{y} - X\vec{\beta})^T V^{-1} (\vec{y} - X\vec{\beta})$ $\nwarrow w_i = \frac{1}{\text{Var}(y_i)}$ ideally

• correlated errors:

• V no longer diagonal

• normal equations: $X^T V^{-1} X \hat{\beta} = X^T V^{-1} y$

• problem: need to estimate V ; see later

• Tests & Confidence intervals:

• Regression params:

$e_i \sim N(0, \sigma^2)$ $\swarrow$ (j,j) element of $(X^T X)^{-1}$

$\hat{\beta}_j \sim N(\beta_j, \sigma^2 v_j)$

$\frac{RSS}{\sigma^2} \sim \chi^2_{N-p}$

$\hat{\sigma}^2 = \frac{RSS}{N-p}$

$\Rightarrow t = \frac{\hat{\beta}_j - \beta_j}{\sqrt{\hat{\sigma}^2 v_j}} \sim t_{N-p}$ $se(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 v_j}$

$100(1-\alpha)\% \text{ CI} = \hat{\beta}_j \pm t_{n-p, \frac{\alpha}{2}} \cdot se(\hat{\beta}_j)$

• Test hypothesis of the form $\beta_j = \beta^*$ for some β^* ,
 eg $H_0: \beta_j = 0$

$$T = \frac{\hat{\beta}_j}{\text{se}(\hat{\beta}_j)} \sim t_{N-p} \text{ under } H_0$$

• Linear combination of regression params:

$$\psi = \tilde{z}^T \beta, \quad \hat{\psi} = \tilde{z}^T \hat{\beta}$$

$$\hat{\psi} \sim N(\psi, \sigma^2 v) \quad \text{where } v = \tilde{z}^T (X^T X)^{-1} \tilde{z}$$

$$\frac{\hat{\psi} - \psi}{\sqrt{\hat{\sigma}^2 v}} \sim t_{N-p}$$

leaving max
q non-zero

• Tests about multiple params

• Test subset $p-q$ params are all 0,
 H_1 : at least 1 of $p-q$ params in H_0 isn't 0

RSS_0, RSS_1 are RSS under H_0, H_1

$$\frac{RSS_0 - RSS_1}{\sigma^2} \sim \chi^2_{p-q} \quad \left(\text{as } \frac{RSS_0}{\sigma^2} \sim \chi^2_{N-q}, \frac{RSS_1}{\sigma^2} \sim \chi^2_{N-p} \right)$$

$$\frac{(RSS_0 - RSS_1) / (p-q)}{RSS_1 / (N-p)} \sim F_{p-q, N-p} \text{ under } H_0$$

Linear Models - Multiple Linear Regression

$$Y_i = \vec{x}_i^T \vec{\beta} + e_i$$

$$\hat{\beta} = (X^T X)^{-1} X^T Y, \quad \hat{\beta} \sim N(\vec{\beta}, \sigma^2 (X^T X)^{-1})$$

$$RSS = (\vec{Y} - X^T \hat{\beta})^T (\vec{Y} - X^T \hat{\beta}) = \vec{Y}^T \vec{Y} - \hat{\beta}^T X^T \vec{Y}$$

$$\sigma^2 = \frac{RSS}{N-m-1}$$

$$H_0: \beta_j = 0 \Rightarrow t = \frac{\hat{\beta}_j}{\text{se}(\hat{\beta}_j)} \sim t_{N-m-1}$$

$H_0: \forall$ of parameters are all 0

$$\Downarrow$$
$$F = \frac{(RSS_0 - RSS)/p}{RSS/(N-m-1)} \sim F_{p, N-m-1}$$

Interpretation of Regression params:

- Partial regression coefficient:

" β_j in multiple linear regression model measures rate of change in $\vec{x}_j$ with all others fixed"
mean response when with

- Total regression coefficient:

- when only 1 response variable, β_j is total regression coef

- measure rate of change of mean response with $\vec{x}_j$, ignoring other explanatory variables

- Partial & total regression coeffs only equal if X-matrix columns are orthogonal

- Checking model adequacy:
 - Checking assumptions:

X	1	2	3
	X ₁		
		X ₂	
			Y

- Matrix plot:

- predictors vs predictors:

check collinearity & leverage points not violations of assumptions, but problematic

- predictors vs response:

check linearity, outliers & homoscedasticity

- Residuals & Standardized results

$$\hat{e}_i = (Y_i - \hat{X}_i^T \hat{\beta}) \text{ estimators for } e_i$$

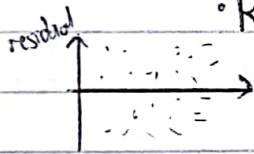
standardized residuals

$$r_i = \frac{\hat{e}_i}{\hat{\sigma} \sqrt{1 - h_{ii}}} \quad (H = X(X^T X)^{-1} X^T)$$

r_i should have mean 0, skew 1; assess their size (~5% larger than 2)

- Residual plots:

random pattern = good



- Predictor vs residuals: autocorrelation non-linearity, heteroscedasticity
- Fitted val vs residuals: same as above
- Observation order vs residuals: informative, eg time (if order is informative) autocorrelation, heteroscedasticity

- Normal probability plots

should look like a straight line

indicate ~~no~~ deviating from normality & outliers

- Remedies for violated assumptions:

- Non-linearity:

transform predictors (eg log, exp...) or non-linear models

- Non-normality: (help skew)

transformation, robust regression (esp help outliers)

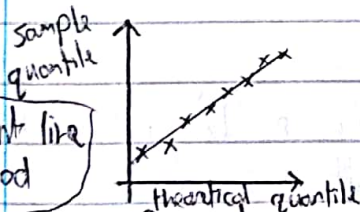
- Heteroscedasticity:

weighted least squares, robust regression if only a few outliers

- Dependence (of errors):

possibly time series models

only really effects stdevs & CIs



straight line = good

- R^2 , coefficient of determination

$$R^2 = \frac{\sum (\hat{y}_i - \bar{y})^2}{\sum (y_i - \bar{y})^2} = 1 - \frac{RSS}{CTSS} \quad \leftarrow CTSS = \sum (y_i - \bar{y})^2$$

- proportion of total variation explained by model
- small = assumption violated, or predictors missing or large error variance

Outliers:

- Regression outliers: large residuals on plots
- Leverage points: potential of a single point to influence model
 - don't violate model assumptions
 - good: if 'in-line' w/ other points, ie omitting this point won't change model much
 - bad: if out of line, removing point changes model a lot

standardised measure of leverage $\rightarrow MD_i = \sqrt{(\vec{x}_i - \bar{x})^T \hat{\Sigma}_x^{-1} (\vec{x}_i - \bar{x})}$; $MD_i^2 = (N-1) \left[h_{ii} - \frac{1}{N} \right]$

- Cook's Statistic: $(MD_i^2 \sim \chi^2_{p-1})$
 $\hat{\beta}_{(i)}$ is estimator of $\vec{\beta}$, omitting point i

$$D_i = \frac{1}{p\sigma^2} (\hat{\beta} - \hat{\beta}_{(i)})^T X^T X (\hat{\beta} - \hat{\beta}_{(i)})$$

$$\left(= \frac{1}{p} \left(\frac{h_{ii}}{1-h_{ii}} \right) r_i^2 \right) \leftarrow \text{efficient} \quad \leftarrow \text{(residual)}$$

Problems:

- $X^T X$ non-invertible; Cook's statistic unstable as $\hat{\beta}$ unstable
- Masking effect:
 - leverage points 'hide' each other; need to use robust estimators, can then look at residual plots to help find outliers

Linear Models - Robust Regression

- Outliers are common in real world data, often as ^{some} explanatory variables are inevitably missed
 - may want to ignore them or not (eg Finance)
- In LS regression, squared residuals make sensitive to outliers (ie try to minimise $\hat{\sigma}^2$)
 - Instead, try to minimise absolute ^{sum} or median residual (ie try to minimise $\hat{\sigma}$)

- Effect of Leverage points:
 - (slope estimator) $\hat{\beta}_1 = \frac{\sum (x_i - \bar{x}) y_i}{\sum (x_i - \bar{x})^2} = \sum v_i y_i$ ($v_i = \frac{x_i - \bar{x}}{\sum (x_k - \bar{x})^2}$)
 - in simple linear regression,
 - largest weight to large $(x_i - \bar{x})$ points (ie large leverage)
 - R^2 analogy $\Rightarrow$ good leverage $\uparrow R^2$
bad leverage $\downarrow R^2$

- Efficiency:
 - designed to be less affected by outliers
 - $\Rightarrow$ asymptotic variance higher than that of LS estimator

M-estimation:

- minimise sum of objective function, ρ , or residuals
 - $\sum \rho(r_i)$, where $\rho(u) \geq 0$, $\rho(0) = 0$, $\rho(u) = \rho(-u)$
 $\rho(u) \geq \rho(u')$ if $|u| \leq |u'|$

$\hat{\beta}_M$ minimises $\sum \rho\left(\frac{e_i(\vec{\beta})}{\sigma}\right) = \sum \rho\left(\frac{Y_i - \vec{x}_i^T \vec{\beta}}{\sigma}\right)$

$\Rightarrow \sum \vec{x}_i \rho'\left(\frac{Y_i - \vec{x}_i^T \hat{\beta}_M}{\sigma}\right) = 0 = \sum \vec{x}_i w_i u_i$

robustness weights $w_i = \frac{\rho'(u_i)}{u_i}$
 $u_i = (Y_i - \vec{x}_i^T \hat{\beta}_M) / \sigma$

σ is unknown:
estimate w/

$\sigma_{MAD} = \text{MED}(|\hat{r}_i - \text{MED}(\hat{r}_j)|)$
(need to iterate this too)

if w_i fixed, $\hat{\beta}_M = (X^T W X)^{-1} X^T W Y$ ($W = \text{diag}(w_1, \dots, w_n)$)
as not, iterate $\hat{\beta}_M^{(k+1)} = (X^T W^{(k)} X)^{-1} X^T W^{(k)} Y$ until convergence

- residuals only enter through $\rho' \Rightarrow$ bounded $\rho' =$ bounded effect of large residuals

• Distribution of M-estimators & efficiency:

• Consistent if $E(\rho'(Z)) = 0$, where $Z \sim F$

• holds for all symmetric F if ρ' bounded

• Tests & CIs:

$$\sqrt{n}(\hat{\beta}_n - \beta) \overset{d}{\sim} N(0, V(\rho', F)L^{-1}), \quad V(\rho', F) \text{ complicated matrix}$$

$$L = \lim_{n \rightarrow \infty} \frac{1}{n} X^T X$$

• Efficiency:

$$C = \sigma^2 (X^T X)^{-1} \quad \text{for LS}$$

$$V(\psi, F)L^{-1} = bC \quad \text{for M-estimation}$$

$1/b = \text{efficiency}$, need $b \times$ as many observations to match LS estimator precision

• Objective function:

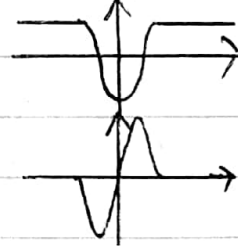
$$\rho(u) = u^2 \Rightarrow \text{LS}$$

$$\rho(u) = |u| \Rightarrow L_1 \text{ - Regression}$$

$$\rho_B(u) = \begin{cases} \frac{1}{6} (1 - [1 - (\frac{u}{c})^2]^3) & \text{if } |u| \leq c \\ \frac{1}{6} & \text{if } |u| > c \end{cases}$$

$$\rho'_B(u) = \begin{cases} \frac{u}{c^2} [1 - (\frac{u}{c})^2]^2 & \text{if } |u| \leq c \\ 0 & \text{if } |u| > c \end{cases}$$

ρ' not small for large residuals; unbounded
not robust to leverage points



Bisquare

0 for large residuals!

c tunes efficiency vs robustness

• S-estimator:

• Bisquare M-estimator relies on (unknown) σ

M-estimator of scale $\rightarrow M(\vec{Z}) = \frac{1}{N} \sum_i \rho\left(\frac{z_i}{S_M(\vec{Z})}\right) = E_{N(0,1)} \rho(Z)$ for observations $(\vec{z}_1, \dots, \vec{z}_N)$

$$\rho(u) = u^2 \Rightarrow E_{N(0,1)} \rho(Z) = 1 \quad \text{and} \quad S_M^2(\vec{Z}) = \frac{1}{N} \sum_i z_i^2$$

• Linear regression:

minimize $S_M(e_1(\vec{\beta}), \dots, e_N(\vec{\beta})) = S_M$
 $= \min S_M(e_1(\vec{\beta}), \dots, e_N(\vec{\beta}))$

residuals

• Computation:

- find $\beta^{(k+1)}$ for given $S_M^{(k)}(\vec{z}^{(k)})$ & $\beta^{(k)}$
- find $S_M^{(k+1)}$ for given $\vec{z}^{(k+1)} = (e_1(\beta^{(k+1)}), \dots, e_N(\beta^{(k+1)}))$
- repeat until convergence
- maybe local optima; repeat w/ diff $\beta^{(0)}$

- MM-estimator:

- combine S & M-estimation

- S_M estimator of σ^2 , then use this for M-estimator
iterate

Linear Models - Variable Selection

$$Y_i = \vec{x}_i^T \vec{\beta} + e_i$$

- aim to reduce #no. variables, ie find $\beta_i = 0$ (or close to 0)
- motivation:
 - + p large, N small $\Rightarrow X^T X$ nearly colinear, unstable estimation^{1A}
 - $N > 10p$ desired
 - + simpler, clear explanation
 - worse for prediction
 - unstable variable selection if some are correlated
 - bias remaining β to be high, so p-values biased low
- t-tests & F-tests:
 - F-test anova table: no variable affects response
 - F-test: y of $\beta_i = 0$ (given other $m-1$ in model)
 - t-test: $\beta_i = 0$, given other $m-1$ in model
 - note: dropping 1 variable changes p-values for others!
- Best subset selection:
 - m variables $\Rightarrow 2^m$ possible models; which is best?
 - for fixed k no. variables, find model which:
 - minimizes $RSS, \hat{\sigma}^2$, maximizes R^2
 - cannot use this to compare models w/ different k
- Criteria to compare models:
 - Akaike Information Criteria (AIC)
 - $AIC = -2\hat{\lambda}(\text{model}) + 2p$ (no. regression params (=d+1))
 - $\hat{\lambda}(\text{model})$ is max of (log-likelihood function)
 - $\hat{\lambda}(\text{model}) = -\frac{N}{2} \log \hat{\sigma}^2 - \frac{RSS}{2\hat{\sigma}^2} + \text{const.}$
 - $\Rightarrow AIC = N \log \frac{RSS}{N} + 2p$, using $\hat{\sigma}_{ML}^2 = \frac{RSS}{N}$
 - Other functions of RSS :
 - Mallows C_p or adjusted R^2 ; in general too 'soft'; doesn't reduce no. vars enough

Leave-one-out
cross validation



• LOO-CV:

- train on $N-1$ points, test on holdout point
- repeat N times & average losses
- computationally demanding;
- + not based on any model assumptions

• Stepwise methods:

- best subset is exponential in no. variables; too slow!

• Backward elimination:

- fit full model

- for $k=m$ to 0:

- fit ~~model~~ all models w/ $k-1$ vars (k dif models)

- find best

- get sequence of 'best' models for all k , use AIC or LOO-CV to pick best from sequence

• Forward selection

- Start from 0 variables, keep adding variable which produces largest decrease in RSS

- again select from sequence produced w/ AIC or LOO-CV

- can use A/F-tests here to decide when to stop

- backwards/forwards give dif models; often backwards better

- not guaranteed to find best subset model, & very unstable

• Lasso:

- standardise variables (to mean 0, stdev 1)

$$x_{ij} = \frac{Z_{ij} - \bar{z}_j}{s_j}, \text{ minimise } S(\vec{\beta}) = \sum e_i^2$$

$$\text{subject to } \sum_{i=1}^m |\beta_i| \leq t$$

makes β_i
comparable size

- set t by LOO-CV

- encourages sparse solutions, & equivalent to L_1 -regularization

+ more stable & 'smooth' for large m & close-to collinearity situations

- biased, worse than stepwise in 'clear-cut' situations

- computation is complicated

Linear Models: ANOVA ← Analysis of Variance

- Y_i depends on categorical variables $\in \{0, 1\}$
- ANOVA tries to breakdown of $\frac{RSS}{CTSS}$ into sum of RSS & SS

- Comparison of Groups - One Way Layout
 $Y_{ij} = \mu_i + e_{ij}, \quad i = 1, \dots, I \leftarrow \text{groups}$
 $j = 1, \dots, n_i$

$$\vec{Y} = X\vec{\beta} + \vec{e}$$

$$\vec{Y} = \begin{pmatrix} Y_{11} \\ \vdots \\ Y_{In_I} \end{pmatrix}, \quad X = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}, \quad \vec{\beta} = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_I \end{pmatrix}, \quad \vec{e} = \begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix}$$

$\xrightarrow{\quad I \quad}$

• cols of $X = I$ indicator variable, rows = n observations

• eg $n_1 = 2, n_2 = 3, n_3 = 1 \Rightarrow X = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

• this way, $\vec{\beta} = \vec{\mu}$

• usual hypothesis: $\mu_1 = \mu_2 = \dots = \mu_I: H_0, H_1: \text{at least 2 differ}$

• test with $RSS_0 = CTSS$

- Two Way Layout:

• observations classified according to 2 factors

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + e_{ijk}$$

$\uparrow$ overall expected value $\uparrow$ effect of level i of factor 1 $\uparrow$ effect of level j of factor 2 $\uparrow$ interaction effect

$i = 1, \dots, I, j = 1, \dots, J$
 $k = 1, \dots, n_{ij}$

• for $I = J = n_{ij} = 2$

$$Y = \begin{pmatrix} Y_{111} \\ Y_{112} \\ Y_{121} \\ Y_{122} \\ \vdots \\ Y_{222} \end{pmatrix}, \quad X = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}, \quad \vec{\beta} = \begin{pmatrix} \mu \\ \alpha_1 \\ \alpha_2 \\ \beta_1 \\ \beta_2 \\ \gamma_{11} \\ \gamma_{12} \\ \gamma_{21} \\ \gamma_{22} \end{pmatrix}$$

• but $X^T X$ not invertible as too many params

• need to introduce constraints, eg

$\sum \alpha_i = 0, \sum \beta_j = 0$ and $\sum \gamma_{ij} = 0, \sum_j \gamma_{ij} = 0$
 (allow one of each to be determined by all others, effectively allowing us to ~~remove~~ shrink X)

• these lead to

$$X = \begin{pmatrix} 1 & -1 & -1 & -1 \\ 1 & -1 & -1 & -1 \\ 1 & -1 & -1 & -1 \\ 1 & -1 & -1 & -1 \\ 1 & -1 & -1 & -1 \\ 1 & -1 & -1 & -1 \\ 1 & -1 & -1 & -1 \\ 1 & -1 & -1 & -1 \\ 1 & -1 & -1 & -1 \\ 1 & -1 & -1 & -1 \end{pmatrix}, \vec{\beta} = \begin{pmatrix} \mu \\ \alpha_2 \\ \beta_2 \\ \gamma_{22} \end{pmatrix} \quad (\text{other params from constraints})$$

• Usual F-tests: (or AIC/LOO-CV)

no interactions $\rightarrow$ all $\gamma_{ij} = 0$
can remove a factor $\rightarrow$ all $\alpha_i = 0$
 $\rightarrow$ all $\beta_j = 0$

done in order is a
sort of 'restricted back selection'

Generalised Linear Models - General Theory

Components:

- Random $Y_1, \dots, Y_N$
- Systematic η_i
- Link function $g(\mu_i) = \eta_i, \mu_i = E(Y_i)$ (monotonic & differentiable)

Linearity:

$$\eta_i = \vec{x}_i^T \vec{\beta}$$

Special cases:

• Linear Models: $g(\mu) = \mu$ (identity link)

• Logistic model: $g(\mu) = \log\left(\frac{\mu}{1-\mu}\right) \leftarrow n=1 \Rightarrow \text{logit of } \mu$

• used for Bernoulli trials ($n=1$) data, or $Y_i \sim \text{Binomial}(n_i, \pi_i)$

• Log-linear model:

$$Y_i \sim \text{Poisson}(\mu_i) \Rightarrow \log(\mu_i) = \vec{x}_i^T \vec{\beta} \Rightarrow g(\mu) = \log(\mu)$$

• Contingency tables:

Exponential family:

$$f(y; \theta, \phi) = \exp\left[\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi)\right]$$

↑
dispersion/scale param

$a(\phi) = \phi/w$ for known w

Distribution	θ	ϕ	$b(\theta)$	$c(y, \phi)$
Poisson(μ)	$\log \mu$	1	e^θ	$-\log y!$
Bin(n, π)	$\log \frac{\pi}{1-\pi}$	1	$n \log(1+e^\theta)$	$\log \binom{n}{y}$
N(μ, σ^2)	μ	σ^2	$\frac{1}{2} \theta^2$	$-\frac{1}{2} (y^2/\phi + \log(2\pi\phi))$

(for all, $w=1$)

• known $\phi \Rightarrow$ exp family

• unknown $\phi \Rightarrow$ exp dispersion family

• Properties:

$$E(Y) = b'(\theta)$$

$$Var(Y) = b''(\theta) a(\phi)$$

$$Var(Y) = V(\mu) a(\phi) = V(\mu) \frac{\phi}{w}$$

• Canonical links:

• g such that $g(\mu) = \theta$

• log, logit & identity for Poisson, Bin & N

differentiate $\int f(y) dy = 1$,
use $a(\phi) \neq 0$

$$f(y_i; \theta_i, \phi) = \exp \left[\frac{y_i \theta_i - b(\theta_i)}{a_i(\phi)} + c(y_i, \phi) \right]$$

• Some GLM theory:

- Y_i from distribution w/ pdf in exp family (& independent)
- let $E(Y_i) = \mu_i$, $\eta_i = \sum_j \beta_j x_{ij}$
- Estimation:

$$\cdot l = \sum l_i, \quad l_i = \log f(y_i; \theta_i, \phi) [= \log P(y)]$$

$$\Rightarrow \frac{\partial l}{\partial \beta_j} = \sum_i \left(\frac{y_i - \mu_i}{V(\mu_i)} \right) \frac{d\mu_i}{d\eta_i} x_{ij} = 0$$

• Iterative procedure:

• Estimation of β :

• cannot solve likelihood eqns algebraically

• Fisher scoring method:

• initialize $\beta^{(0)}$,

• iterate

$$\rightarrow \underbrace{(X^T W X)^{(s-1)}}_{\text{evaluated w/ } \hat{\beta}^{(s-1)}} \hat{\beta}^{(s)} = (X^T W \tilde{z})^{s-1}$$

$$w_{ii} = \frac{1}{V(\mu_i)} \left(\frac{d\mu_i}{d\eta_i} \right)^2$$

$$z_i = \eta_i + (y_i - \mu_i) \left(\frac{d\eta_i}{d\mu_i} \right)$$

• Estimation of ϕ :

Pearson
Statistic

$$\rightarrow X^2 = \sum_i \left[\frac{(y_i - \hat{\mu}_i)^2}{V(\hat{\mu}_i)} \right] \quad 0 \text{ mean, unit variance RVs}$$

$$\frac{X^2}{\phi} \sim \chi^2_{N-p} \Rightarrow \hat{\phi} = \frac{\hat{X}^2}{N-p}$$

• Sampling distribution of $\hat{\beta}$:

Taylor expand
around true β_0 ,
evaluate at $\hat{\beta}$

$$\rightarrow \frac{\partial l}{\partial \beta} \Big|_{\hat{\beta}} \approx \frac{\partial l}{\partial \beta} \Big|_{\beta_0} + (\hat{\beta} - \beta_0) \frac{\partial^2 l}{\partial \beta^2} \Big|_{\beta_0}$$

$$\Rightarrow (\hat{\beta} - \beta_0) \approx - \frac{\frac{\partial l}{\partial \beta} \Big|_{\beta_0}}{\frac{\partial^2 l}{\partial \beta^2} \Big|_{\beta_0}} \sim N(0, I^{-1}) \quad \text{as } n \rightarrow \infty \text{ (by CLT)}$$

$$\Rightarrow \hat{\beta} \sim N_p(\vec{\beta}, I^{-1})$$

($I = X^T W X / \phi$, for W at convergence of Fisher Scoring method)

GLM - General Theory

Fisher information matrix
(j, k)th element

(ie the -Hessian!)

Covariance Matrix of $\hat{\beta}$:

$$E\left(-\frac{\partial^2 \ell}{\partial \beta_j \partial \beta_k}\right) = \sum_i \sum_{i'} \left(\frac{E[(Y_i - \mu_i)(Y_{i'} - \mu_{i'})]}{\text{Var}(Y_i) \text{Var}(Y_{i'})} \right) \frac{\partial \mu_i}{\partial \eta_i} \frac{\partial \mu_{i'}}{\partial \eta_{i'}} x_{ij} x_{i'k}$$

$$\mathcal{I} = \sum_i \left(\frac{x_{ij} x_{ik}}{\text{Var}(Y_i)} \right) \cdot \left(\frac{\partial \mu_i}{\partial \eta_i} \right)^2 \begin{pmatrix} \text{as} \\ E[(Y_i - \mu_i)(Y_i - \mu_i)] \\ \text{Var}(Y_i) \text{ if } i=i' \\ \text{Cov}(Y_i, Y_{i'})=0 \text{ if } i \neq i' \end{pmatrix}$$

$$\mathcal{I} = X^T W X / \phi$$

Covariance matrix of $\hat{\beta}$ is $\mathcal{I}^{-1} = (X^T W X)^{-1} \phi$

Confidence intervals for model parameters:

- assume known dispersion parameter, ϕ (unknown ϕ same, except must estimate ϕ & use appropriate t-distribution)
- Single parameter:

$$\hat{\beta} \sim \text{approx } N_p(\vec{\beta}, \mathcal{I}^{-1})$$

approx (1- α)% CI $\rightarrow \Rightarrow \hat{\beta}_j \pm z_{\alpha/2} \text{se}(\hat{\beta}_j)$, where $\text{se}(\hat{\beta}_j) = \sqrt{(\mathcal{I}^{-1})_{jj}}$

linear combination of parameters:

$$\psi = \vec{c}^T \hat{\beta}$$

$$\psi \sim \text{approx } N(\psi, \vec{c}^T \mathcal{I}^{-1} \vec{c})$$

Hypothesis testing:

$H_0: \beta_j = 0 \Rightarrow \frac{\hat{\beta}_j}{\text{se}(\hat{\beta}_j)} \sim N(0, 1)$ under H_0

likelihood test

H_0 : subset of $p-q$ params all 0

$$\Rightarrow 2[\ell(\hat{\beta}) - \ell(\hat{\beta}_0)] \sim \chi^2_{p-q} \text{ under } H_0 \text{ (applies when } \phi \text{ known)}$$

Similar to RSS

Deviance:

$$D = 2[\ell(\hat{\beta}_{\text{sat}}) - \ell(\hat{\beta})] \phi \quad \left(\begin{array}{l} \ell(\hat{\beta}_{\text{sat}}) \text{ is saturated likelihood,} \\ \text{achieved when } \hat{\mu}_i = y_i \end{array} \right)$$

scaled deviance

$$\rightarrow D^* = D/\phi \sim \chi^2_{N-p} \text{ (under } H_0 \text{ above)}$$

can also repress likelihood test

$$D_0^* - D^* \sim \chi^2_{p-q}$$

• unknown ϕ :

• under same H_0 , $D_0^* - D^* \sim \chi^2_{p-q}$ & $D^* \sim \chi^2_{N-p}$

• if $(D_0^* - D^*) \perp D^*$, under H_0 & in large sample limit

$$F = \frac{(D_0^* - D^*) / (p-q)}{D^* / (N-p)} = \frac{(D_0 - D) / (p-q)}{\underbrace{D / (N-p)}_{\text{no need to know } \phi}} \sim F_{p-q, N-p}$$

GLM - Binomial Data & Logistic Regression

$Y_1, \dots, Y_N$ independent, $Y_i \sim \text{Bin}(n_i, \pi_i)$

$$g(\mu_i) = \vec{x}_i^T \vec{\beta}, \quad \mu_i = E(Y_i) = n_i \pi_i$$

$$\eta = \vec{x}_i^T \vec{\beta} = \log\left(\frac{\pi_i}{1-\pi_i}\right)$$

logit link gives linear logistic model (CDF of logistic dist) $\pi = \frac{1}{1+e^{-\eta}}$

(alternatively: probit (inverse standard normal CDF) $\pi = \Phi(\eta)$)
complementary log-log $(\vec{x}_i^T \vec{\beta} = \log[-\log(1-\pi_i)])$ (CDF of extreme value dist) $\pi = 1 - \exp(-e^\eta)$

• these choices ensure $0 \leq \pi \leq 1$ & $-\infty < g(\mu) < \infty$
($\Rightarrow$ no constraints on $\vec{\beta}$)

• Interpretation of logistic regression parameters:

• $\text{logit}(\pi) = \log\left(\frac{\pi}{1-\pi}\right) = \text{log-odds}$

• β_j measures rate of change of log odds w/ x_j (8 other x_i held constant)

• eg $x_i \in \{0, 1\} \Rightarrow$ odds on success w/ $x_i = 1$ is odds of success w/ $x_i = 0 \times \exp[\beta_i]$

• Some basic results:

• likelihood: $\frac{\partial \ell}{\partial \beta_j} = \sum_i \frac{y_i - \mu_i}{V(\mu_i)} \cdot \frac{\partial \mu_i}{\partial \eta_i} x_{ij} = \sum_i (y_i - \hat{\mu}_i) x_{ij} = 0$

• MLE:

$$(X^T W X)^{(s-1)} \hat{\beta}^{(s)} = (X^T W^{(s)} \vec{z})^{(s-1)}$$

• for logistic model, $W_{ii} = \frac{1}{V_i} \left(\frac{\partial \mu_i}{\partial \eta_i}\right)^2$

$$W_{ii} = n_i \pi_i (1 - \pi_i)$$

$$\left(\begin{aligned} (V_i = \text{Var}(Y_i) = V(\mu_i)) \\ = n_i \pi_i (1 - \pi_i) \\ \frac{d\mu_i}{d\eta_i} = n_i \pi_i (1 - \pi_i) \end{aligned} \right)$$

$$\vec{z}_i = \eta_i + (y_i - \mu_i) \left(\frac{d\eta_i}{d\mu_i}\right), \quad \left(\frac{d\eta_i}{d\mu_i} = \frac{1}{n_i \pi_i (1 - \pi_i)}\right)$$

• Sampling distribution of $\hat{\beta}$:

$$\hat{\beta} \overset{\text{approx}}{\sim} N_p(\vec{\beta}, (X^T W X)^{-1}) \quad \text{for large } N$$

Fisher Iterations

$$\text{Deviance: } D = 2 \sum_i \left[y_i \log\left(\frac{y_i}{\hat{\mu}_i}\right) + (n_i - y_i) \log\left(\frac{n_i - y_i}{n_i - \hat{\mu}_i}\right) \right]$$

$D \sim \chi^2_{N-p}$ if model is true

$H_0: \beta_j = 0$

$$\Rightarrow \frac{\hat{\beta}_j}{\text{se}(\hat{\beta}_j)} \sim N(0,1)$$

$H_0: \forall \text{ of } \beta_{1, \dots, \beta_m} \text{ are } 0$

$$\Rightarrow D_0 - D \sim \chi^2_v$$

alternative
goodness of
fit test

Pearson Chi-Square statistic:

$$\chi^2 = \sum_i \frac{(y_i - n_i \hat{\pi}_i)^2}{n_i \hat{\pi}_i (1 - \hat{\pi}_i)} \sim \chi^2_{N-p} \text{ under model}$$

(can be shown $D \approx \chi^2$)

$(\chi^2 = \sum \frac{\overset{\text{observed}}{O} - \overset{\text{expected}}{E}}{E})^2$

Binary data:

if $n_i = 1 \forall i$, results still hold but D no longer good for assess quality of fit

can test by grouping observations into $g \sim 10$ groups according to $\hat{\pi}_j$, then calculate $\chi^2 \sim \chi^2_{g-2}$

Checking model adequacy:

Plot residuals:

• Raw

• Pearson

• Standardized Pearson

• Deviance

• Standardized Deviance

• Cooks statistic:

$$\hat{e}_i = y_i - n_i \hat{\pi}_i$$

$$X_i = \hat{e}_i / \sqrt{n_i \hat{\pi}_i (1 - \hat{\pi}_i)}$$

$$r_{P_i} = X_i / \sqrt{1 - h_{ii}}$$

$$\sum d_i^2 = D$$

$$r_{D_i} = d_i / \sqrt{1 - h_{ii}}$$

$$D_i = \frac{1}{p} \left(\frac{h_{ii}}{1 - h_{ii}} \right) r_{P_i}^2$$

$$\begin{pmatrix} h_{ii} = H_{ii} \\ H = W^{1/2} X (X^T W X)^{-1} X^T W^{1/2} \end{pmatrix}$$

GLM - Binomial Data & Logistic Regression

- Model selection:

- Stepwise / best subset selection:

- can then use AIC, which for binomial data

$$AIC = D + 2p$$

- Compare model deviances:

$$D_{M_1} - D_{M_2} \sim \chi^2_{p_1 - p_2} \quad (\text{equivalent to likelihood test})$$

- Analysis of deviance:

GLM - Contingency tables: log-linear modelling

A \ B	B ₁	...	B _J
	n ₁₁	...	n _{1J}
A _i			
A _j	n _{j1}		n _{jJ}

π_{ij} = prob observation in cell (i,j)

$\mu_{ij} = N \pi_{ij}$ = expected frequency

if $A \perp\!\!\!\perp B$, $\pi_{ij} = \pi_{i+} \cdot \pi_{+j}$, $\mu_{ij} = \frac{\mu_{i+} \cdot \mu_{+j}}{N}$

expected row frequency
↓
expected column frequency

$H_0: A \perp\!\!\!\perp B \Rightarrow \log \mu_{ij} = \log \mu_{i+} + \log \mu_{+j} - \log N$

$H_1: A \not\perp\!\!\!\perp B \Rightarrow \log \mu_{ij} = \alpha_i + \beta_j + \lambda + \phi_{ij}$
(= $A + B + A:B$)

log-linear models
(log of expected response)

this is over-parameterised; need constraints on parameters to solve

eg $\alpha_i = \beta_j = \phi_{ij} = \phi_{i1} = 0$

or $\sum_i \alpha_i = \sum_j \beta_j = \sum_{ij} \phi_{ij} = 0$

H_1 is saturated; as many ~~free~~ effective params as table cells,
 $\Rightarrow$ can fit perfectly, $N \pi_{ij} = n_{ij}$

no clear interpretation of params; mainly want to test if zero

Three-way contingency tables:

now variables A, B, C w/ I, J, K categories

Saturated model:

$$\log \mu_{ijk} = A + B + C + A:B + A:C + B:C + A:B:C = A * B * C$$

again, over-parameterised, need similar constraints to reduce to IJK effective parameters

Model Submodels:

1. $A + B + C$

2. $A + B + C + A:B$

3. $A + B + C + A:B + A:C$

4. $A + B + C + A:B + A:C + B:C$

Independences

$A \perp\!\!\!\perp B \perp\!\!\!\perp C$

$A, B \perp\!\!\!\perp C$

$B \perp\!\!\!\perp C | A$

π_{ijk}

$\pi_{i++} \cdot \pi_{+j+} \cdot \pi_{++k}$

$\pi_{ij+} \cdot \pi_{++k}$

$\pi_{ij+} \cdot \pi_{i+k} / \pi_{i++}$

$\uparrow \quad \uparrow \quad \uparrow$
 $P(B, C | A) \quad P(C | A) \quad P(A)$

$P(B, C | A) \cdot P(A)$

- Fitting the models:
 - maximal likelihood (joint dist of observation is multinomial)
 - equivalent to joint distribution of independent Poisson RVs, conditioned on their sum
 - $\Rightarrow$ fit as if observed frequencies are independent Poisson RVs

1 • Goodness of fit & model checking:

- done as for Poisson data, using G^2

$$G^2 = 2 \sum o \log \left(\frac{o}{e} \right) \quad (o = \text{observed}, e = \text{expected})$$

- can compare nested models using G^2

• Residuals:

- Raw

$$\hat{e}_{ij} = n_{ij} - \hat{\mu}_{ij}$$

- Pearson

$$X_{ij} = \hat{e}_{ij} / \sqrt{\hat{\mu}_{ij}}$$

- Standardised Pearson

$$r_{p,ij} = X_{ij} / \sqrt{1 - h_{ij}}$$

- Deviance

$$\sum_{ij} d_{ij}^2 = D$$

- Standardised Deviance

$$r_{d,ij} = d_{ij} / \sqrt{1 - h_{ij}}$$

Generalized Additive Models: An Overview

$$g[E(Y_i)] = \eta_i = \vec{X}_i^* \vec{\Theta} + f_1(x_{i1}) + f_2(x_{i2}) + f_3(x_{i3}, x_{i4}) + \dots$$

$\vec{X}_i^*$ = i^{th} row of X^*

X^* = model matrix for any parametric model components

$\vec{\Theta}$ = parameter vector

f_j = smooth function for covariates x_{ij}

note $\sum_j f_j(x_{ij}) = 0$

Difference w/ GLM:

GLM models effect of x_{ij} as $\Theta_0 + \overbrace{\sum_j \Theta_j x_{ij}}^{\text{linear!}}$

GAM models

as $\sum_j f_j(x_{ij})$

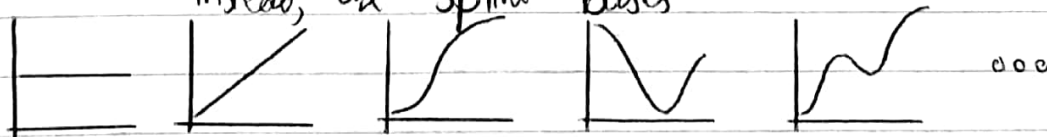
Smooth functions:

represented using regression splines

$$f_j(x_j) = \sum_k \beta_{jk} b_{jk}(x_j), \text{ where } b_{jk} \text{ are basis functions \& } \beta_{jk} \text{ regression parameters to learn}$$

basis function could be polynomial ($x, x^2, x^3, \dots$), but these become highly collinear

instead, use spline bases



Parameter estimation:

estimated using MLE, adding smoothing parameters $\vec{\lambda}$ to reduce overfitting (choose $\vec{\lambda}$ by, eg, LOO-CV)

$$\begin{aligned} \text{maximize } l(\vec{\beta}) - \frac{1}{2} \sum_j \lambda_j \int [f_j^{(d_j)}(x_j)]^2 dx_j \\ = l(\vec{\beta}) - \frac{1}{2} \sum_j \lambda_j \vec{\beta}^T S_j \vec{\beta} \end{aligned}$$

derivative of f (d_j usually = 0)
encourage less rough functions

Don't estimate $\vec{\lambda}$; would just choose very low regularisation!

known coefficients matrix $\rightarrow S = \sum_j \lambda_j S_j$

$$\Rightarrow \hat{\beta} = (X^T W X + J)^{-1} X^T W \vec{z}$$

(note biased estimator,
due to penalty)

• Inference:

• Inference not standard due to fixed λ_j
 $\vec{\beta} | \vec{y} \sim N(\hat{\beta}, (I + J)^{-1})$ allows approximate inference
 (eg CIs) to be done

• Variable selection & model comparison also not standard